

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Code of Conduct

I Pratip raj /192521409(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Pratip raj P

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Hospital information system for SIMATS Hospital using any Cloud Service Provider to demonstrate SaaS with fields of patient name, age, address, phone, email, Attender name, attender phone, doctor name, diseases, etc.

Experiment 2: Create a simple cloud software application for online super market using any Cloud Service Provider to demonstrate SaaS with the template of customer name, address, phone, email, items, quantity etc. with the item classification form.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 3 CPU, 10GB RAM and 10GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

Assessment Tasks

Experiment 1: Create a simple cloud software application for Mark Sheet Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as name of the student, Reg no of the student , Subjects , Marks, Average, percentage, rank, overall performance etc.

Experiment 2: Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Buy, Rent, commercial, Rental Agreement , Rental Agreement, Area, place, Range, property value etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 5 CPU, 12 GB RAM and 8 GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM.

19/08/2016

Cloud Computing Big Data Analysis Assessment Tool-2

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192521409

- ① Create a simple cloud software application for SIMATS hospital using any cloud service provider to demonstrate SaaS with fields

Aim:

To create a simple cloud based hospital information system using a cloud service provider and demonstrate the software as a service (SaaS) model

Software / requirements:

- * Cloud service provider
- * web browser
- * internet connection
- * HTML, CSS and JavaScript
- * cloud-based hosting platform

Fields used:

The hospital application contains the following fields

Field

Example

Patient name

Prathip Raj

Age

20

Address

Chennai

Phone

987651104

Email

Patient@gmail.com

Doctor name

Dr. Kumar

Disease

Fever

Procedure :

Step 1: open the selected Cloud Service Provider
Create / sign in to your account

Step 2: Create a new web application / Project
named hospital information System

Step 3: design a simple hospital registration Form

Step 4: Add the required input Field;

Patient name, age, address, phone, email, attender name
Doctor name, disease

Step 5: add a Submit / Register button to the form

Step 6: Connect the application to cloud storage / database
if provided by the selected cloud platform

Step 7: Deploy / host the application on the cloud

Step 8: Submit the details and verify that the
information is stored / displayed correctly

Sample output:

SIMATS Hospital Information System

Patient name : Prastip
age : 20
address : Chennai
Phone : 9876543210
email : Prastip@gmail.com
Attender name : Raj kumar
Doctor name : Dr. Kumar
Disease : Fever

[Submit]

and
Result:

The cloud based hospital information system was successful created and deployed.

② Create a simple cloud software application for an online supermarket using any application for an online supermarket using any cloud service provider to demonstrate SaaS.

Aim:

To create a simple cloud-based online supermarket application using a cloud service provider and demonstrate the software as a service.

Software requirements:

Cloud service provider, web browser, internet connection, internet connection, HTML, CSS and JavaScript, Cloud-based hosting/database

Field used:

- * Customer name - royal.lk
- * Address - Chennai
- * Phone - 9876543210
- * Email - Customer@gmail.com
- * item name - rice
- * item classification - Grocery
- * Quantity - 2

- Procedure:
- * open the selected cloud service provider and Sign
 - * Create a new project named online Supermarket
 - * Design a Customer and Shopping Form
 - * add fields for Customer name, address, Phone and email
 - * add a Submit/Order button
 - * Connect the application to cloud storage / database if available
 - * open the application using a web browser
 - * enter Customer and item details and Submit the Form

Sample output:

ONLINE SUPERMARKET
Customer order Form

Customer Name	: Ragul. bk
address	: Chennai
Phone	: 9876543210
email	: Customer@gmail.com
item name	: Rice
item Classification	: Grocery
Quantity	: 2

[Place order]

Result:

The Cloud based online Supermarket application was successfully created and deployed. It allows customers to enter their details, select items.

Sign

1) Demonstrate virtualization by installing a Type-2 hypervisor, creating and configuring a vm image with windows/Linux as the host os and create a virtual machine with 3 CPU, 10 GB RAM

Aim:

To demonstrate virtualization by installing a Type 2 hypervisor and creating a virtual machine with the required CPU, RAM and storage configuration

Software requirements:

host operating system windows / Linux, Type 2 hypervisor, virtual machine ISO Image, Computer / Laptop

VM Configuration:

Resource

Configuration

CPU

-

3 CPU

RAM

-

10 GB

Storage

10 GB

Host OS

-

windows / Linux

virtualization

-

Type-2

Procedure:

* start the computer with windows / Linux as the host os

* install a Type-2 hypervisor

- * open the hypervisor
- * select the Create / new virtual machine
- * enter the VM name
- * Configure the processor to 3 CPU
- * Allocate 10 GB RAM
- * Create a virtual hard disk with 10 GB Storage
- * Complete the VM Creation
- * Verify the CPU, RAM and Storage Configuration

Sample output:

Virtual machine

VM Name : Test Vm
 Host OS : windows / Linux
 CPU : 3 CPU
 RAM : 10 GB
 Storage : 10 GB
 Status : Running

Concept:

A Type 2 hypervisor runs on top of an existing host operating system. It allows multiple virtual machines to be created and managed on the same physical computer.

Result:

The virtual machine was successfully created using a Type-2 hypervisor with 3 CPU, 10 GB RAM and 10 GB Storage.

Demonstrate virtualization by installing a Type-2 hypervisor. Creating and Configuring a VM image
Creating a Snapshot of VM and testing it by loading the Previous Version / Cloned VM

Aim:

To demonstrate the Snapshot Feature of virtualization^{ion} by creating a Snapshot of a Virtual machine and restoring / testing the previous version

Software requirements:

Windows / Linux host OS, Type 2 hypervisor, Virtual machine, guest operating system, Sufficient Storage

Procedure:

- * start the Computer with windows / Linux as the host OS
- * open the installed Type-2 hypervisor
- * Create or open a virtual machine
- * install and configure the guest operating system
- * open the Snapshot option in the hypervisor

- * Create a new Snapshot
- * Give the Snapshot a Suitable name, Such as before Changes
- * make Some Changes to the virtual machine
- * Shut down or pause the VM
- * Select the previously Created Snapshot
- * restore/load the Snapshot
- * Start the VM and verify that the previous VM state has been restored

Sample output:

VM Snapshot

VM Name : Test VM

Snapshot Name : before changes

initial State : working

Change made : Yes

Snapshot Created : Yes

Restore Snapshot : successful

Previous version : loaded

Status : Tested Successfully

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-pt:

A VM Snapshot Saves the state of a virtual machine at a particular point in time. it can be used to return the VM to that earlier state after changes have been made

result:

The VM Snapshot was successfully created and tested. The previous version of the virtual machine was successfully loaded.